

Data Communication

Content 09: Cellular Communication

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Cellular Communication

- **Definition:** Cellular communication divides a large geographic area into smaller regions called **cells**, each served by a base station.
- **Purpose:** To enable **wireless voice and data communication** for mobile users while efficiently reusing frequencies.
- **How it works:**
 - Mobile users communicate with the nearest base station.
 - Calls or data sessions are transferred (handoff) as users move between cells.
 - Frequency reuse increases system capacity.
- **Evolution:**
Started with analog 1G → digital 2G (GSM/CDMA) → 3G → 4G (LTE) → 5G.
- [Useful Youtube Link](#)

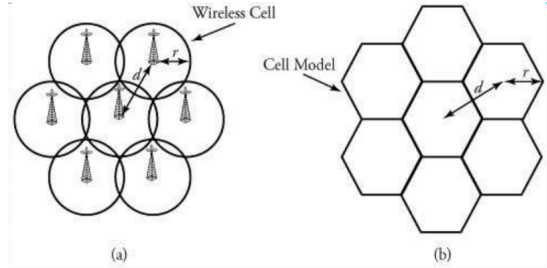


Cell

- A **cell** is the basic geographical coverage area in a cellular network.
- Each cell is controlled by a **Base Station (BS)** equipped with antennas and transceivers.
- Cells are often represented as hexagons for simplicity in planning.
- Each cell uses a specific set of frequencies to avoid interference with adjacent cells.



Cell



- Hexagon is a better choice for cell because they can cover the geographic region with the fewest number of cells rather than square or triangle.
- Hexagon also closely approximates a circular radiation pattern which would occur for an Omnidirectional antenna.

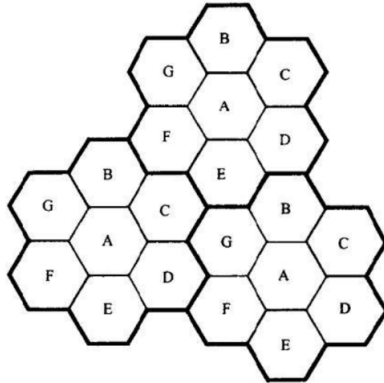
Frequency Reuse

- Base stations in adjacent cells are assigned channel groups which is completely different than neighboring cells.
- The base station antennas are designed to achieve the desired coverage within a particular cell.
- By limiting the coverage area within the boundaries of a cell, the same group of channels may be used to cover different cell that are separated from one another by keeping the interference level within tolerable range. This is called **Frequency Reuse**.



Frequency Reuse

The cells which collectively use the complete set of available frequencies is called a **Cluster**.



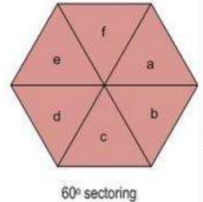
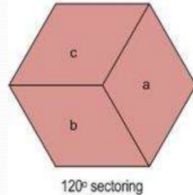
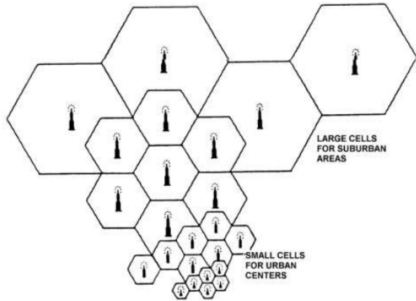
Handoff

- When a mobile moves into a different cell while a conversion is in progress, the MSC automatically transfers the call to a new channel belonging to the new base station. This is called handoff.
- **Types:**
 - **Hard Handoff:** “Break before make”
 - Existing link is broken before a new one is made.
 - Risk of call drop but simpler implementation.
 - Used in GSM.
 - **Soft Handoff:** “Make before break”
 - New link is established before breaking the old one.
 - More reliable, especially for fast-moving users.
 - Used in CDMA.
- Ensures seamless communication without call drops.



Capacity Improvement Techniques

- As user demand increases, capacity must be improved without increasing spectrum.
- Key strategies:
 - **Cell Splitting:** Dividing large cells into smaller ones.
 - **Sectoring:** Dividing a cell into 3 or 6 directional sectors using directional antennas.



General Architecture

MS $\rightarrow$ BTS (in one cell) $\rightarrow$ MSC $\rightarrow$ BTS (in another cell) $\rightarrow$ MS

Where:

MS = Mobile System (User Device)

BTS = Base Transceiver Station (each covers one cell)

MSC = Mobile Switching Centre (routes call/data to destination)

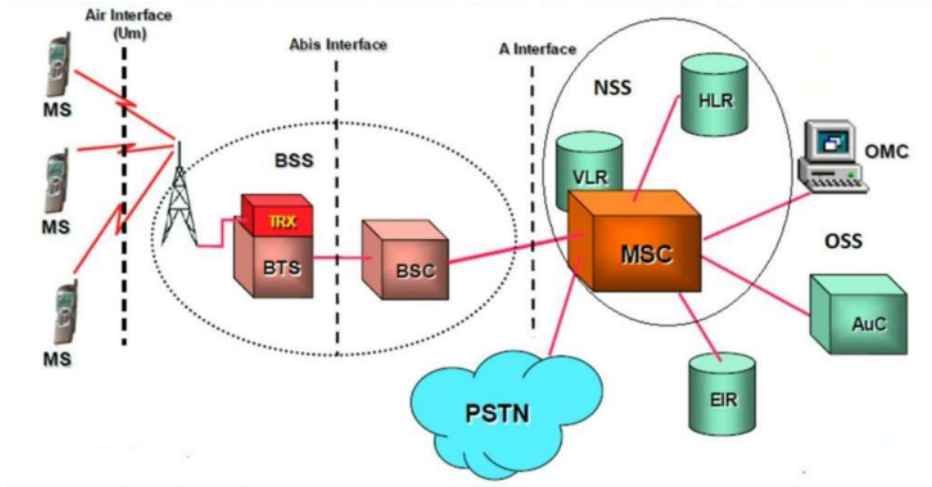


GSM Architecture

- GSM (Global System for Mobile Communication) is a 2G cellular standard.
- **Main Components:**
 - **Mobile Station (MS)** – handset + SIM card.
 - **Base Station Subsystem (BSS)** – handles radio communication.
 - **Network Switching Subsystem (NSS)** – call routing, authentication.
 - **Operation and Support Subsystem (OSS)** – maintenance and control.
- Uses **TDMA (Time Division Multiple Access)** for channel access.



GSM Architecture



GSM Architecture: Terminologies

	Full Form	Function
MS	Mobile Station	User Phone + SIM
BTS	Base Transceiver Station	Handles radio communication
BSC	Base Station Controller	Manages multiple BTS and handoffs
MSC	Mobile Switching Center	Core switch for calls, routing, mobility
HLR	Home Location Register	Permanent Subscriber Database
VLR	Visitor Location Register	Temporary user data for roaming
AuC	Authentication Center	Verifies SIM identity
EIR	Equipment Identity Register	Tracks IMEI, blocks stolen devices
OMC	Operation & Maintenance Center	Controls configuration, alarms
PSTN	Public Switched Telephone Network	Connects to landline



GSM Architecture: How It Works

- MS → BTS (Air Interface – UM)
The Mobile Station (MS) sends a call request to the Base Transceiver Station (BTS) over the air interface.
- BTS → BSC (Abis Interface)
The BTS forwards this request to the Base Station Controller (BSC), which allocates radio channels and manages handoffs between BTSs.
- BSC → MSC (A Interface) The BSC sends the call setup request to the Mobile Switching Center (MSC) through the A interface.
- MSC and Databases (HLR, VLR, AuC, EIR) The MSC verifies subscriber information and call permissions by checking:
 - VLR (Visitor Location Register): Temporary user info
 - HLR (Home Location Register): Permanent subscriber data
 - AuC (Authentication Center): Validates SIM credentials
 - EIR (Equipment Identity Register): Checks handset IMEI validity

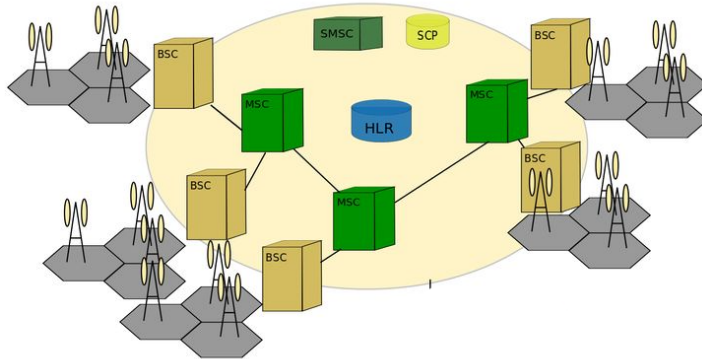


GSM Architecture: How It Works

- MSC → PSTN / Other MSC
After authentication, the MSC routes the call — either to another mobile user (via another MSC) or to a landline through the Public Switched Telephone Network (PSTN).
- During the Call: The BSC and BTS maintain the radio link. If the user moves, handoff ensures the call continues smoothly.

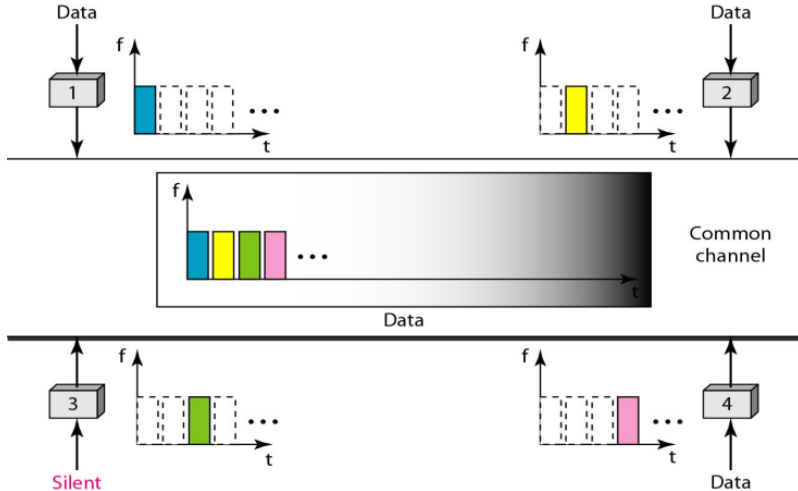


IS-95 Architecture

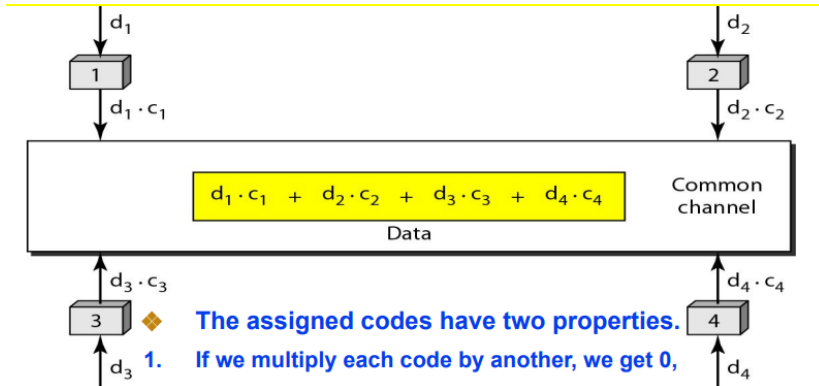


Uses **CDMA (Code Division Multiple Access)** for channel access.

In Depth: TDMA



In Depth: CDMA



The assigned codes have two properties.

1. If we multiply each code by another, we get 0,
2. If we multiply each code by itself, we get 4
(the number of stations)

$$\text{Data} = (d_1 c_1 + d_2 c_2 + d_3 c_3 + d_4 c_4) \times c_1 = 4 \times d_1$$

IS-95 (CDMA) vs GSM

- **Access Method:**
 - GSM uses TDMA.
 - CDMA uses code-based multiplexing.
- **Handoff:** GSM uses hard handoff; CDMA uses soft handoff.
- **Efficiency:** CDMA is more spectrum-efficient but complex.
- **Voice Quality:** CDMA generally offers better quality in low-signal areas.
- **Adoption:** GSM became globally dominant due to open standardization.



Why CDMA Failed

- CDMA was technologically superior but faced practical and economic challenges:
 - Proprietary technology – limited openness and licensing restrictions.
 - Difficult international roaming and interoperability.
 - GSM's early adoption created strong network effects.
 - Upgrades to 3G and 4G favored GSM-based evolution (UMTS, LTE).
- Ultimately, CDMA declined as GSM evolved into LTE and 5G standards.



Thank You

